ELE 8125 - Signals & Systems

Lecture 4 - Part 2

Fourier Series representation of Continuous-Time Periodic Signals

Linear Combinations of Harmonically Related Complex Exponentials

Periodic signal: $x(t) = x(t+T)$ $\forall t$

$x(t) = \cos(\omega_0 t)$ (sinusoidal signal)

$x(t) = e^{j\omega_0 t}$ (periodic complex exponential)

$\Rightarrow$ fundamental frequency ω_0

$\Rightarrow$ fundamental period $T = 2\pi/\omega_0$

set of harmonically related complex exponentials: $\boxed{\varphi_k(t) = e^{jk\omega_0 t} = e^{jk(2\pi/T)t}, \quad k = 0, \pm 1, \pm 2, \ldots}$

linear combination of harmonically related complex exponentials: $\boxed{x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}}$

- $k = 0$, $x(t)$ constant $\Rightarrow$ DC
- $k = 1$ & $k = -1$, have frequency ω_0 $\Rightarrow$ "first harmonic components"
- $k = 2$ & $k = -2$, "second harmonic components"
- $\vdots$
- $k = N$ & $k = -N$, "N-th harmonic components"

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

$x(t)$ could be complex $\Rightarrow$ Assuming $x(t) = x^*(t)$ $\Rightarrow$ $x(t)$ is real $\Rightarrow$ $x(t) = x^*(t) = \left(\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}\right)^* = \sum_{k=-\infty}^{\infty} a_k^* e^{-jk\omega_0 t}$

since $x(t)$ is real $\Rightarrow$ $a_k = a_{-k}^*$ $\Rightarrow$ $\boxed{x(t) = \sum_{k=-\infty}^{\infty} a_{-k}^* e^{jk\omega_0 t}}$ (replacing k by -k)

$a_k^* = a_{-k}$

$$x(t) = \sum_{k=-\infty}^{\infty} a_{-k}^* e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$



when:
- $k = 0 \Rightarrow a_0 e^{j(0)\omega_0 t} = a_0$
- $k = 1 \Rightarrow a_1 e^{j\omega_0 t}$
- $k = -1 \Rightarrow a_{-1} e^{-j\omega_0 t}$
- $k = 2 \Rightarrow a_2 e^{j2\omega_0 t}$
- $k = -2 \Rightarrow a_{-2} e^{-j2\omega_0 t}$
- $\vdots$
- $k = l \Rightarrow a_l e^{jl\omega_0 t}$
- $k = -l \Rightarrow a_{-l} e^{-jl\omega_0 t}$

to simplify the summation and taking only the positive values of k $\Rightarrow$ $\left(a_k e^{jk\omega_0 t} + a_{-k} e^{-jk\omega_0 t}\right)$

$$\Rightarrow x(t) = a_0 + \sum_{k=1}^{\infty} \left(a_k e^{jk\omega_0 t} + \underbrace{a_{-k}}_{a_k^*} e^{-jk\omega_0 t}\right) \Rightarrow x(t) = a_0 + \sum_{k=1}^{\infty} \left(a_k e^{jk\omega_0 t} + a_k^* e^{-jk\omega_0 t}\right) = a_0 + \sum_{k=1}^{\infty} \left[a_k e^{jk\omega_0 t} + (a_k e^{jk\omega_0 t})^*\right]$$

$(a+jb)+(a+jb)^* = a+jb+a-jb \Rightarrow 2a \Rightarrow \boxed{(a+jb)+(a+jb)^* = 2a}$

$$x(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k e^{jk\omega_0 t} + (a_k e^{jk\omega_0 t})^*\right] = a_0 + \sum_{k=1}^{\infty} Re\{2a_k e^{jk\omega_0 t}\} \Rightarrow \boxed{x(t) = a_0 + 2\sum_{k=1}^{\infty} Re\{a_k e^{jk\omega_0 t}\}}$$

a_k is complex $\Rightarrow$ expressed in polar form $\Rightarrow$ $a_k = A_k e^{j\theta_k}$

$$\Rightarrow x(t) = a_0 + 2\sum_{k=1}^{\infty} Re\{A_k e^{j\theta_k} e^{jk\omega_0 t}\} = a_0 + 2\sum_{k=1}^{\infty} Re\{A_k e^{j(k\omega_0 t+\theta_k)}\}$$

$$A_k e^{j(k\omega_0 t+\theta_k)} = A[\cos(k\omega_0 t+\theta_k) + j\sin(k\omega_0 t+\theta_k)] = A_k\cos(k\omega_0 t+\theta_k) + jA_k\sin(k\omega_0 t+\theta_k)$$

$$\Rightarrow Re\{A_k e^{j(k\omega_0 t+\theta_k)}\} = Re\{A_k\cos(k\omega_0 t+\theta_k) + jA_k\sin(k\omega_0 t+\theta_k)\} = A_k\cos(k\omega_0 t+\theta_k)$$

$$\Rightarrow \boxed{Re\{A_k e^{j(k\omega_0 t+\theta_k)}\} = A_k\cos(k\omega_0 t+\theta_k)}$$

$$\Rightarrow \boxed{x(t) = a_0 + 2\sum_{k=1}^{\infty} A_k\cos(k\omega_0 t+\theta_k)}$$

Common form of the Fourier series of real periodic signals in continuous time

Writing a_k in rectangular form $\Rightarrow a_k = B_k + jC_k$

$$\Rightarrow x(t) = a_0 + 2\sum_{k=1}^{\infty} Re\{a_k e^{jk\omega_0 t}\} = a_0 + 2\sum_{k=1}^{\infty} Re\{(B_k+jC_k)e^{jk\omega_0 t}\} = a_0 + 2\sum_{k=1}^{\infty} Re\{B_k e^{jk\omega_0 t} + jC_k e^{jk\omega_0 t}\}$$

$$= a_0 + 2\sum_{k=1}^{\infty}\left(Re\{B_k e^{jk\omega_0 t}\} + Re\{jC_k e^{jk\omega_0 t}\}\right)$$

$$Re\{B_k e^{jk\omega_0 t}\} = B_k Re\{\cos(k\omega_0 t) + j\sin(k\omega_0 t)\} = B_k\cos(k\omega_0 t) \Rightarrow \boxed{Re\{B_k e^{jk\omega_0 t}\} = B_k\cos(k\omega_0 t)}$$

$$Re\{jC_k e^{jk\omega_0 t}\} = C_k Re\{j(\cos(k\omega_0 t) + j\sin(k\omega_0 t))\} = C_k Re\{-\sin(k\omega_0 t) + j\cos(k\omega_0 t)\} = -C_k\sin(k\omega_0 t)$$

$$\Rightarrow \boxed{Re\{jC_k e^{jk\omega_0 t}\} = -C_k\sin(k\omega_0 t)}$$

$$\Rightarrow \boxed{x(t) = a_0 + 2\sum_{k=1}^{\infty}\left[B_k\cos(k\omega_0 t) - C_k\sin(k\omega_0 t)\right]}$$

Three equivalent forms of Fourier Series in terms of complex exponentials for real periodic functions

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\left(\frac{2\pi}{T}\right)t}$$

$$x(t) = a_0 + 2\sum_{k=1}^{\infty} A_k\cos(k\omega_0 t+\theta_k)$$

$$x(t) = a_0 + 2\sum_{k=1}^{\infty}\left[B_k\cos(k\omega_0 t) - C_k\sin(k\omega_0 t)\right]$$

Synthesis Equation: $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$

Analysis Equation: $a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t}\,dt = \frac{1}{T}\int_T x(t) e^{-jk(2\pi/T)t}\,dt$, a_0: dc component

$$a_0 = \frac{1}{T}\int_T x(t)\,dt$$

Example: Consider the signal: $x(t) = \sin(\omega_0 t)$

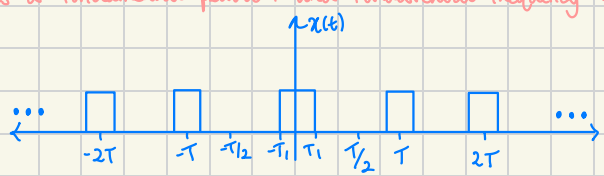
$$\sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} = \frac{1}{2j}e^{j\omega_0 t} - \frac{1}{2j}e^{-j\omega_0 t}$$

By identification: $a_1 = \frac{1}{2j}$, $a_{-1} = \frac{-1}{2j}$, $a_k = 0$ since $k \neq +1$ or -1

Example 2: The periodic square wave, sketched in the figure below and defined over one period is:

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

The signal has a fundamental period T and fundamental frequency $\omega_0 = 2\pi/T$



$$a_0 = \frac{1}{T}\int_T x(t)\,dt = \frac{1}{T}\int_{-T_1}^{T_1} dt = \frac{1}{T}\left[T\right]_{-T_1}^{T_1} = \frac{1}{T}\left[T_1 + T_1\right] = \frac{2T_1}{T} \Rightarrow \boxed{a_0 = \frac{2T_1}{T}}$$

$$a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t}\,dt = \frac{1}{T}\int_{-T_1}^{T_1} e^{-jk\omega_0 t}\,dt \quad , \quad [-jk\omega_0 t]' = -jk\omega_0 \Rightarrow 1 = \frac{1}{-jk\omega_0}[-jk\omega_0 t]'$$

$$= \frac{1}{T}\int_{-T_1}^{T_1} \frac{1}{-jk\omega_0}[-jk\omega_0 t]' e^{-jk\omega_0 t}\,dt = \frac{-1}{jk\omega_0 T}\int_{-T_1}^{T_1} [-jk\omega_0 t]' e^{-jk\omega_0 t}\,dt = \frac{-1}{jk\omega_0 T}\left[e^{-jk\omega_0 t}\right]_{-T_1}^{T_1}$$

$$= \frac{-1}{jk\omega_0 T}\left[e^{-jk\omega_0 T_1} - e^{jk\omega_0(T_1)}\right] = \frac{1}{k\omega_0 T}\underbrace{\left[\frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{(2)j}\right]}_{\sin(k\omega_0 T_1)}(2) = \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{2\sin(k\omega_0 T_1)}{k\left(\frac{2\pi}{T}\right)T} = \frac{\sin(k\omega_0 T_1)}{k\pi}$$

$$\Rightarrow \boxed{a_k = \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{\sin(k\omega_0 T_1)}{k\pi}}$$